

Chapter 1 Introduction to vibrations

1. RMS 与 MS

R: root M: mean S: square

2. $\hat{x}$ 表示(实数)幅值

3. 单弹簧系统

$$\omega = \sqrt{\frac{k}{m}}$$

4. 阻尼系统

$$-kx - \beta \frac{dx}{dt} = m \frac{d^2x}{dt^2}$$

$$\text{阻尼项} -\frac{\beta}{2m} = A$$

$$\text{总: } e^A e^{\pm \sqrt{\omega_0^2 - A^2} t} \quad \omega_0 \text{ 为无阻尼时系统固有频率 } \omega_0 = \sqrt{\frac{k}{m}}$$

5. 受迫振动

$$-kx - \beta \frac{dx}{dt} + f(t) = m \frac{d^2x}{dt^2} \quad (1.1)$$

注意外力正方向与位移同向,
回复力和阻力与位移反向

$$\text{设 } f(t) = \hat{f} e^{j\omega t} \quad x(t) = \hat{x} e^{j(\omega t + \phi)}$$

① 求解幅值

$$-k\hat{x}e^{j\omega t} - j\beta\omega\hat{x}e^{j\omega t} + f(t) = -m\omega^2\hat{x}e^{j\omega t}$$

$$\hat{x}e^{j\omega t}(k + j\beta\omega - m\omega^2) = f(t)$$

$$\hat{x}(\omega) = \frac{|f(t)|}{|k - m\omega^2 + j\beta\omega|}$$
$$= \frac{\hat{f}}{\sqrt{(k - m\omega^2)^2 + \beta^2\omega^2}}$$

② 求解幅角

$$\hat{x}e^{j(\omega t + \phi)} = \frac{\hat{f}e^{j\omega t}}{k - m\omega^2 + j\beta\omega}$$

$$\text{令 } t=0 \quad e^{j\phi} = C(k - m\omega^2 - j\beta\omega)$$

$$\Rightarrow \tan\phi = \frac{-\beta\omega}{k - m\omega^2}$$

③ 求解阻抗

要点: 把(1.1)中的 $x(t)$ 都写成 $v(t)$ 的形式, 于是有

$$-\frac{k}{j\omega}v(t) - \beta v(t) + f(t) = j\omega m v(t)$$

$$f(t) = v(t) \left(j\omega m + \frac{k}{j\omega} + \beta \right)$$

$$Z(\omega) = \frac{f}{v} = j\omega m + \frac{k}{j\omega} + \beta$$

④ 求解 Q factor

Q factor 为功率为最大一半时的 $\frac{\omega_0}{\Delta\omega}$, ($\Delta\omega$: 半功率带宽)

$\Delta\omega$ 是因为有两个 ω 满足条件, 以下简称该条件为 Q 点,

$$\text{Q点时 } P = \frac{1}{2} P_{\max} \Rightarrow |Z| = \sqrt{2} |Z_{\min}| = \sqrt{2} |Z(\omega_0)|$$

$$Z_{\min} = Z(\omega_0) = 0 + \beta = \beta$$

意味着 Q 点, $|j\omega m + \frac{k}{j\omega}| = \beta$

$$\Rightarrow |j\omega m - j\frac{k}{\omega}| = \beta$$

$$\omega m - \frac{k}{\omega} = \pm \beta$$

$$\text{① 当 } \omega m - \frac{k}{\omega} = \beta \text{ 时}$$

$$\omega^2 m - \beta\omega - k = 0$$

$$\omega = \frac{\beta \pm \sqrt{\beta^2 + 4km}}{2m}$$

$$\text{舍去负数解 } \omega_1 = \frac{\beta + \sqrt{\beta^2 + 4km}}{2m}$$

$$\text{② 当 } \omega m - \frac{k}{\omega} = -\beta \text{ 时}$$

$$\text{类似地有 } \omega_2 = \frac{-\beta + \sqrt{\beta^2 + 4km}}{2m}$$

$$\Rightarrow \Delta\omega = \frac{\beta}{m}$$

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{m\omega_0}{\beta}$$

6. 本章速记

$$\text{① 单振子系统 } \omega_0 = \sqrt{\frac{k}{m}}$$

$$\text{② 阻尼: } A = -\frac{\beta}{2m}, e^A e^{\sqrt{\omega_0^2 - A^2} t}$$

$$\text{③ 单振子系统阻抗: } j\omega m + \frac{k}{j\omega} + \beta$$

$$Q \text{ factor: } \frac{m\omega_0}{\beta}$$

Chapter 2 Wave equation, vibrating strings and membranes

1. 从最一般的波动方程谈起

① $f = f(x-vt)$

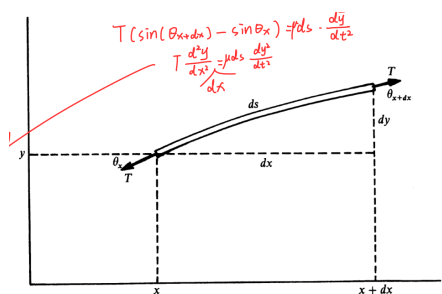
波的本质: 波形整体移动, 在 $x-vt$ 中, 当 t 增大 Δt 时, x 可以同样增大 $v\Delta t$ 而使波形不变, 体现的结果即为波形平移

② 如何确定 v

$$\begin{cases} \frac{d^2 f}{dx^2} = f''(x-vt) \\ \frac{d^2 f}{dt^2} = v^2 f''(x-vt) \end{cases} \Rightarrow \frac{d^2 f}{dt^2} = v^2 \frac{d^2 f}{dx^2}$$

对符合物理现实的模型, 可分别计算 $\frac{d^2 f}{dt^2}$ 和 $\frac{d^2 f}{dx^2}$, 从而确定 v

2. 绳的一维振动的波速



对该情境, 我们有两个近似假设
 $\sin\theta \approx \theta$ $ds \approx dx$

从 $F=ma$ 出发, 对如图的绳微元

$$dF = T \sin(\theta + dx) - T \sin\theta = T \frac{\partial \sin\theta}{\partial x} dx = T \frac{\partial (\frac{\partial y}{\partial x})}{\partial x} dx = T \frac{\partial^2 y}{\partial x^2} dx$$

$$m = \mu ds = \mu dx$$

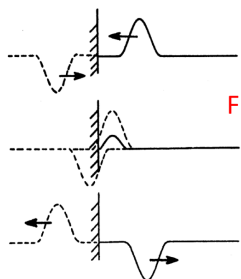
$$\text{于是 } T \frac{d^2 y}{dx^2} dx = \mu \frac{d^2 y}{dt^2} dx$$

$$\text{于是 } v = \sqrt{\frac{T}{\mu}}$$

3. 绳的边界条件

假设除了正向波 $f(x-vt)$, 反射波是从远处而来 $f(x+vt)$

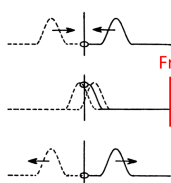
① 固定端



Fixed end
 $y(0,t) = 0$

$$\begin{aligned} f_1(vt) + f_2(-vt) &= 0 \\ f_1(vt) &= -f_2(-vt) \end{aligned}$$

② 自由端



Free end
 $\left[\frac{\partial y(x,t)}{\partial x} \right]_{x=0} = 0$

自由端必平行, 否则绳末端将有无穷大加速度

$$\begin{aligned} f'_1(vt) + f'_2(-vt) &= 0 \\ f'_1(vt) &= -f'_2(-vt) \end{aligned}$$

4. 简谐振动波动方程

$$\tilde{y}(x,t) = \tilde{A}e^{j\omega(x-vt)} + \tilde{B}e^{j\omega(x+vt)}$$

当使用简谐方程时, 往往更关心频率而非波速,

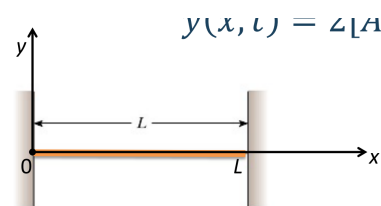
因此更换为 t 本位

$$\tilde{y}(x,t) = \tilde{A}e^{j(\omega t - kx)} + \tilde{B}e^{j(\omega t + kx)}$$

$$\frac{\omega}{k} = v \Rightarrow k = \frac{\omega}{v} \text{ wave number}$$

另一种定义 wave number 的方式是 $k = \frac{2\pi}{\lambda}$

5. 绳上驻波



$$\text{假设波动方程: } y = A \sin(\omega t - kx) + B \cos(\omega t - kx) + C \sin(\omega t + kx) + D \cos(\omega t + kx)$$

$$y(0,t) = 0 \quad A = -C \quad B = -D$$

$$\begin{aligned} \Rightarrow y &= A [\sin(\omega t - kx) - \sin(\omega t + kx)] + B [\cos(\omega t - kx) - \cos(\omega t + kx)] \\ &= 2[A \cos(\omega t) - B \sin(\omega t)] \sin(kx) \end{aligned}$$

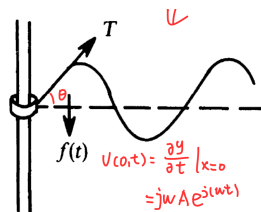
$$y(L,t) = 0 \Rightarrow kL = n\pi$$

$$\Rightarrow \frac{\omega_n}{v} L = n\pi$$

$$\Rightarrow f_n = \frac{\omega_n}{2\pi} = n \frac{v}{2L}$$

6. 绳的输入阻抗

① 自由端



$$f(t) = Fe^{j\omega t}$$

$$\text{自由端只有单向波 } y = \tilde{A}e^{j(\omega t - kx)}$$

$$f(t) = -T \sin\theta$$

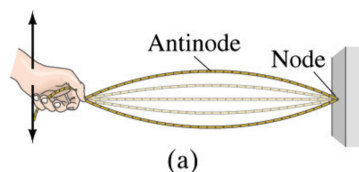
$$= -T \frac{\partial y}{\partial x} \Big|_{x=0}$$

$$= j\omega \tilde{A} k e^{j\omega t}$$

$$V(0,t) = \frac{\partial y}{\partial t} \Big|_{x=0} = j\omega \tilde{A} e^{j\omega t}$$

$$Z_{in} = \frac{f(t)}{V(0,t)} = \frac{-T}{\omega} = -\sqrt{T\mu} \quad (\text{即 } Z_0)$$

② 固定端



固定端引起了反射波.

$$y = \tilde{A}e^{j(\omega t - kx)} + \tilde{B}e^{j(\omega t + kx)}$$

$$y(L,t) = \tilde{A}e^{j(\omega t - kL)} + \tilde{B}e^{j(\omega t + kL)} = 0$$

$$\Rightarrow \tilde{A}e^{jkL} + \tilde{B}e^{-jkL} = 0 \Rightarrow B = -Ae^{-2jkL}$$

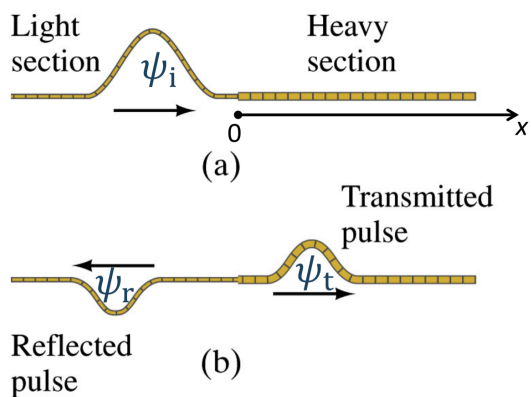
$$V(0,t) = j\omega \tilde{A}e^{j\omega t} + j\omega \tilde{B}e^{j\omega t}$$

$$f(t) = -T \sin\theta = -T(jk\tilde{A}e^{j\omega t} + jk\tilde{B}e^{j\omega t})$$

$$Z_{in} = \frac{f(t)}{V(0,t)} = \frac{-T(jk\tilde{A} - jk\tilde{B})}{j\omega\tilde{A} + j\omega\tilde{B}}$$

$$= Z_0 \frac{1 + e^{-2jkL}}{1 - e^{-2jkL}} = Z_0 \frac{e^{jkL} + e^{-jkL}}{e^{jkL} - e^{-jkL}} = -jZ_0 \cot(kL)$$

7. 阻抗不连续处



一列波 ψ_i 从左侧过来, 到阻抗不连续点, 引起一束反射波 ψ_r , 一束透射波 ψ_t

尽管图上似乎 ψ_i 和 ψ_r 是孤立的, 而且只画了一个波包, 但需要记住的是当 ψ_i 到不连续点处时, 左侧实际的波是 $\psi_i + \psi_r$ 。牢记这一点!

在阻抗不连续点, 两侧的波形可以不连续, 但有些东西必须是连续的

① 位移, 因为绳子没有断开

② 回复力, 因为绳子任一处都不能有无穷大加速度

于是

$$\begin{cases} \psi_L(0,t) = \psi_R(0,t) \\ T_1 \frac{\partial \psi_L(0,t)}{\partial t} = T_2 \frac{\partial \psi_R(0,t)}{\partial t} \end{cases}$$

$$\psi_L = \psi_i + \psi_r \quad \psi_R = \psi_t$$

$$\psi_L = A_i e^{j(\omega t - k_1 x)} + B_r e^{j(\omega t + k_1 x)}$$

$$\psi_R = C_t e^{j(\omega t - k_2 x)}$$

$$\begin{cases} A_i + B_r = C_t \end{cases}$$

$$\begin{cases} T_1 k_1 (-A_i + B_r) = T_2 k_2 C_t \end{cases}$$

$$\begin{cases} A + B = C \end{cases}$$

$$\begin{cases} z_1 A - z_1 B = z_2 C \end{cases}$$

于是

$$\begin{cases} B = \frac{z_1 - z_2}{z_1 + z_2} A \\ C = \frac{2z_1}{z_1 + z_2} A \end{cases}$$

8. 本章速记

① 绳上波速 $c = \sqrt{\frac{T}{\mu}}$

② 波数 $k = \frac{\omega}{v}$

③ 绳上驻波 $\sin(kL) = 0$

④ 自由绳阻抗(特征) $z_0 = \sqrt{T\mu}$

⑤ 固定绳输入阻抗 $z_{in} = z_0 \cot(kL)$

⑥ 阻抗不连续处

$$\begin{cases} \psi_r = R \psi_i \\ \psi_t = T \psi_i \end{cases} \quad \begin{cases} R = \frac{z_1 - z_2}{z_1 + z_2} \\ T = \frac{2z_1}{z_1 + z_2} \end{cases}$$

Chapter 3 propagation of sounds in fluids

1. 基本假设

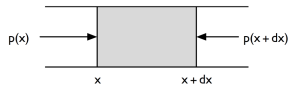
$$\begin{cases} p_{\text{tot}}(\vec{r}, t) = p_0(\vec{r}) + p(\vec{r}, t) \\ \rho_{\text{tot}}(\vec{r}, t) = \rho_0(\vec{r}) + \rho(\vec{r}, t) \end{cases}$$

前一项为静水中的值, 后一项为波传播引起的扰动项

2. 求液体中的波动方程微分关系式

为求得此式, 有三大等式和三大假设

1) 等式1: Force balance



$$dF = dm a$$

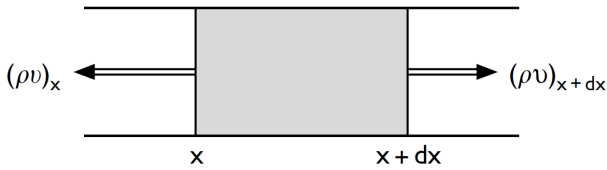
$$\begin{aligned} dF &= (p(x) - p(x+dx))S \\ &= -S \frac{\partial p}{\partial x} dx \end{aligned}$$

$$dm = \rho_{\text{tot}} S dx$$

$$\begin{aligned} a &= \frac{DV_x}{Dt} = \frac{\partial V_x}{\partial t} + V_x \frac{\partial V_x}{\partial x} \quad \text{注意 } a \text{ 是某质点的加速度, 应用物质导数} \\ &= \frac{\partial V_x}{\partial t} + V_x \frac{\partial V_x}{\partial x} \quad \text{为何绳上不考虑此? 因为绳上质点, } x \text{ 坐标不变} \end{aligned}$$

$$\text{联立得 } -\frac{\partial p}{\partial x} = \rho_{\text{tot}} \left(\frac{\partial V_x}{\partial t} + V_x \frac{\partial V_x}{\partial x} \right) \quad (3.1)$$

2) 等式2: mass balance



对图上微元, 近似认为其密度等于左边界密度

$$\frac{d(dm)}{dt} = \frac{d(\rho_{\text{tot}} S dx)}{dt} = \frac{d\rho_{\text{tot}}(x)}{dt} S dx = \frac{d\rho(x)}{dt} S dx$$

同时质量变化也等于流入-流出

$$\begin{aligned} \frac{d(dm)}{dt} &= S (\rho v)_x - (\rho v)_{x+dx} \\ &= -S \frac{\partial(\rho v)}{\partial x} dx \end{aligned}$$

$$-S \frac{\partial(\rho_{\text{tot}} v)}{\partial x} dx = \frac{d\rho(x)}{dt} S dx$$

$$-\frac{\partial(\rho_{\text{tot}} v)}{\partial x} = \frac{d\rho(x)}{dt}$$

$$-\rho_{\text{tot}} \frac{\partial v}{\partial x} - v \frac{\partial \rho_{\text{tot}}}{\partial x} = \frac{\partial \rho}{\partial t} \quad (3.2)$$

3) 等式3: P和\rho的关系

绝热情况下, $p_{\text{tot}} = p_{\text{tot}}(\rho_{\text{tot}})$

$$\text{在 } (p_0, \rho_0) \text{ 处展开 } dp = c^2 d\rho \quad (3.3)$$

4) 假设1: 扰动很小

$$\begin{cases} p(x, t) \ll p_0(x) \\ \rho(x, t) \ll \rho_0(x) \end{cases} \quad (3.4)$$

5) 假设2: 速度很小

$$V_x \approx 0 \quad (3.5)$$

6) 联立

将(3.5)应用到(3.1)和(3.2)上, 消去高阶小量

$$\begin{cases} -\frac{\partial p}{\partial x} = \rho_0 \frac{\partial V_x}{\partial t} \\ -\rho_0 \frac{\partial V_x}{\partial x} = \frac{\partial p}{\partial t} \end{cases} \quad (3.6)$$

将(3.4)应用到(3.6)的第二个式子上, 得

$$\begin{cases} \frac{\partial p}{\partial x} = -\rho_0 \frac{\partial V_x}{\partial t} \\ -\rho_0 \frac{\partial V_x}{\partial x} = \frac{1}{c^2} \frac{\partial p}{\partial t} \end{cases}$$

$$\begin{cases} \frac{\partial^2 p}{\partial x^2} = -\rho_0 \frac{\partial^2 V_x}{\partial t \partial x} \\ -\rho_0 \frac{\partial V_x}{\partial x \partial t} = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} \end{cases}$$

$$\Rightarrow \frac{\partial^2 p}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2}$$

3. 3D waves in fluid

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\nabla^2 p = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2}$$

4. Plane waves

$$\tilde{p}(x, t) = \tilde{A} e^{i(\omega t - kx)} + \tilde{B} e^{i(\omega t + kx)}$$

5. 液体阻抗

$$Z_0 = \rho_0 c$$

6. 声速

$$c = \sqrt{\left| \frac{\partial p}{\partial \rho} \right|_{\text{ad}}} \bigg|_{\rho_0}$$

$$\text{或者引入 } \beta_{\text{ad}} = \frac{1}{\rho_0} \left(\frac{\partial p}{\partial \rho} \right)_0^{\text{(ad)}}$$

$$K_{\text{ad}} = 1/\beta_{\text{ad}}$$

$$c = \sqrt{\frac{K_{\text{ad}}}{\rho_0}}$$

7. 球面波

$$p(r, t) = \frac{1}{r} f(r - ct)$$

8. 本章速记

1) 液体中微分关系式

2) 液体阻抗

3) 液体声速

4) 球面波

